

1 Variants of Batch Grouping

Summary

Batches are grouped and/or combined due to different situations:

- Not relevant to the process
It is assumed here that the identified material only switches positions and this movement is irrelevant to batch tracing (e.g. picking or transfer posting of batches to a group and provision for production).
- Relevant to the process
This function facilitates posting processes allowing to log on many input batches at once to a machine/OP, guide them through the process and to repost them afterwards to produced output batches for traceability purposes.

Variants

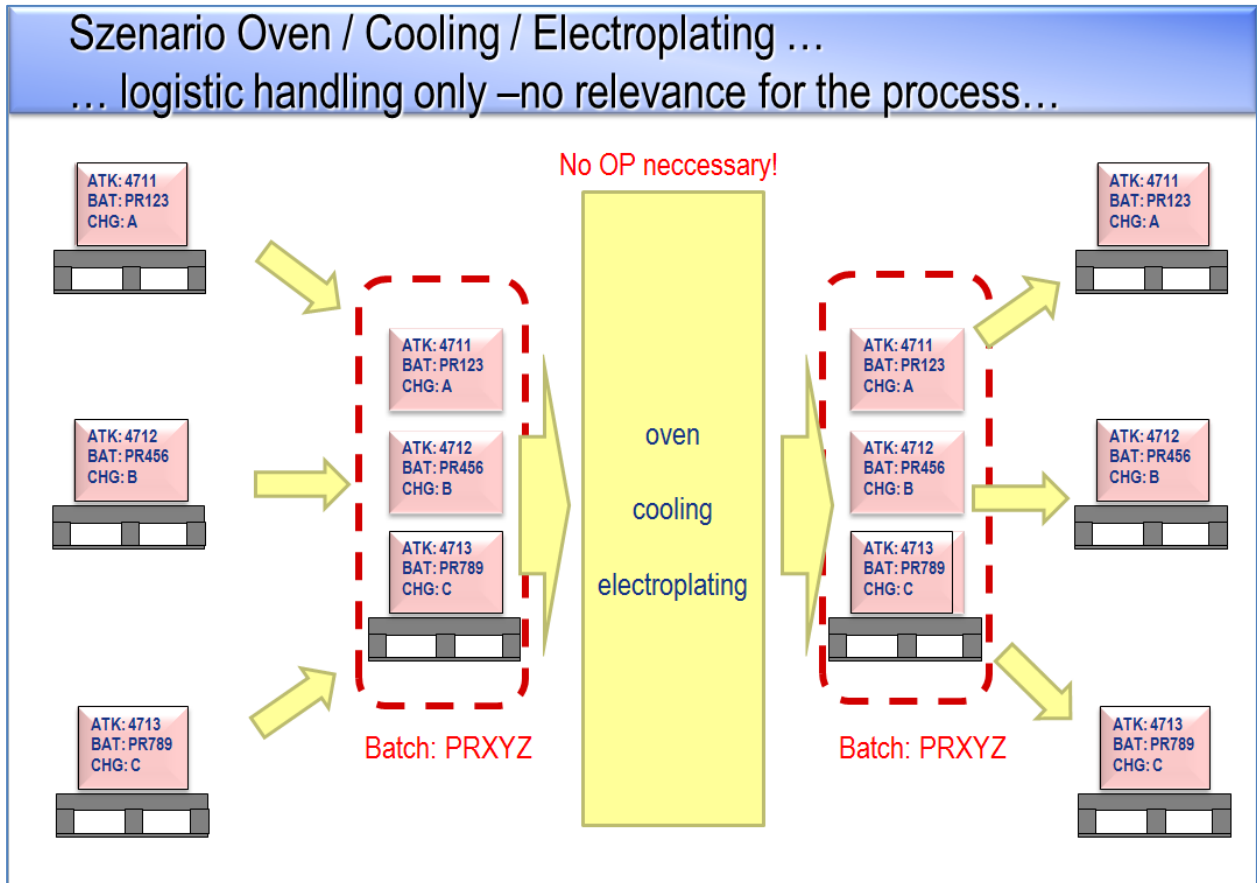
In general, there are the following two variants to group batches.

- Illustration: irrelevant to process --> generate/cancel batch group
- Illustration: relevant to process --> batch grouping

Generate/cancel group batch (not relevant to process)

General

The function is used every time when it is necessary to combine batches of different or identical materials, which is to be resolved/canceled at a later point in time providing the batches "unaffected".



Prerequisites, restrictions

- Original batches remain and are not "consumed" by grouping.
- During grouping, original batches cannot be processed or, for example, archived somewhere else.
- No OP is required - groupings may be performed at any point in time.
- Batches may still be removed unless the grouping has been created finally.
- A label can be printed for the generated grouping.
- Groupings may only be cancelled altogether. Individual batches cannot be removed.
- All group members are affected if the grouping is reposted (change of material buffer).
- Tracing does not consider the grouping.

AIP procedure

There are two steps:

- Generate group batch
- Cancel group batch

Results

The following results can be expected after generating the batch group:

- Batch characteristics
 - Individual batches still keep their material number, material type, material buffer, etc.
- Batch status
 - Once the "generate group batch" function has been executed, the single batches are in the "running" status. The group batch is assigned the "running" status.
- Goods movements
 - No goods movements are posted for the single batches or the group batch.
- Traceability
 - The group of batches is not relevant to tracing.
- History
 - The group of batches is documented in the history of individual batches.

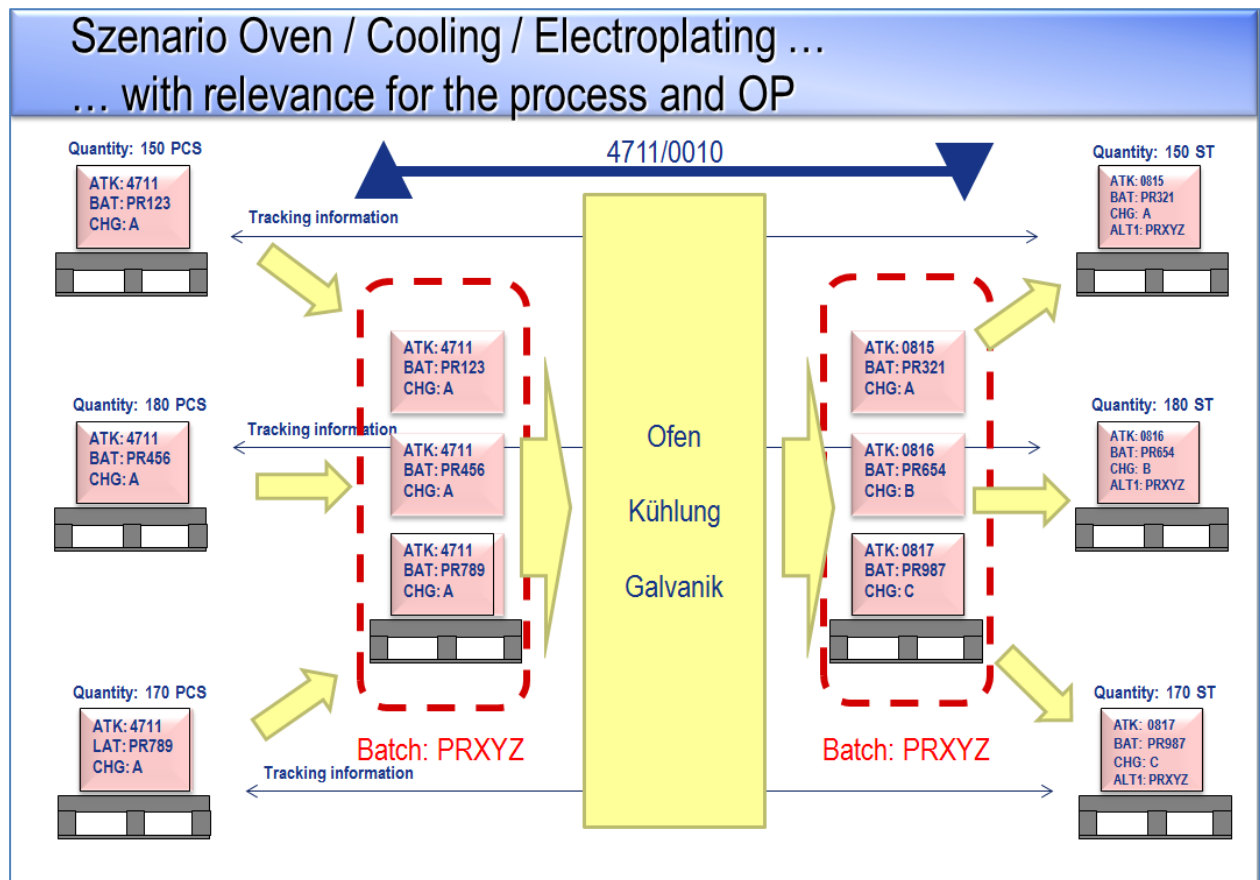
The following results can be expected after canceling the batch group:

- Batch characteristics
 - Individual batches still keep their material number, material type, material buffer, etc.
 - The group batch has taken over the following values from a "system batch" (copy template):
 - Material number
 - Material type
 - Material buffer
 - The quantity of the group batch is the total amount of all assigned individual batches and, therefore, is assigned the unit "PCE/pieces".
- Batch status
 - Once the "cancel group batch" function has been executed, the single batches are in the "free" status. The group batch is assigned the "processed" status. The "grouping" no longer exists.
- Goods movements
 - No goods movements are posted for the single batches or the group batch.
- Traceability
 - Cancelling group batches is not relevant to traceability.
- History
 - Cancelling the group batches is documented in the history of individual batches.

Batch grouping (relevant to process)

General

The function is used every time when batches of different or identical materials are used in a process and output batches are finally changed for all ingoing batches at a point in time.



Prerequisites, restrictions

- This function can only be used if grouping is directly performed "within the process". Groupings cannot be performed beforehand as part of a preceding picking process.
- The machine has not been configured as MPL machine.
- Input batches are "consumed" by grouping.
- During grouping, input batches cannot be processed or, for example, archived somewhere else.
- Operations are required: an OP appropriate for processing is searched via the order network or by reservations affecting all orders.

AIP procedure

The AIP procedure is described here.

Results/ processing

The "batch grouping" dialog (U_GROUPING) provides several functions.

- Add batch, start OP: CNRGRP.ADD
- Remove batch from list: CNRGRP.REMOVE:
- Complete group: CNRGRP.UNLOAD:

After executing the single functions, processing is performed in the dialog U_GROUPING.

Add batch/ start OP (CNRGRP.ADD)

This function adds the entered batch. To do so, the OP for the batch is logged on to the machine.

Inspections:

- General batch inspection:
 - The batch is available
 - The batch has the status "free"
- Automatic search for the operation matching the entered batch
 - OP is searched via the order network and batch assignment
 - The batch was generated in an OP, which is provided as piece of information for the batch. Due to the order network, the system knows the next OP to be processed within the order. The appropriate OP can be found by the order network.
 - If no OP is found via the network, it is searched using batch reservations.
 - An error message appears if no OP is found.
 - If an OP is found, it will only be started, unless it is already running at the machine.
- Further inspections:
 - Checking if the OP is planned for the machine and/or machine group.
 - Checking if the batch is reserved for another OP.
 - Checking if the batch can be logged on to the component.
 - An error message occurs if no component is found.
 - The component must be identical to the material number of the batch, otherwise an error message occurs.
- An error message appears if the admissible number of OPs running at the machine is exceeded.
- Start OP:

- The OP is started and set to the "running" status.
- The batch is reserved for the OP.

Remove batch (CNRGRP.REMOVE):

This function is used to cancel the reservation of the OP at the batch.

Inspections:

- The batch is available. Otherwise, an error message occurs.

Unload batch group (CNRGRP.UNLOAD):

This function is used to log on the reserved batches.

Inspections:

- Are there any running OPs including reserved batches at the machine?

Posting:

- Determination of all reserved batches
- Log output batch and input batch on.
 - At first generate a new batch number for the output batch
 - The new output batch is logged on with the reserved input batch.
- Log off output batch and input batch
 - The input batch is logged off
 - The output batch is logged off
 - The configured, alternative batch numbers and attributes are saved for the output batch.
 - The connection between input batch and output batch is saved for tracing purposes.
- This process is repeated for all batches included in the group.